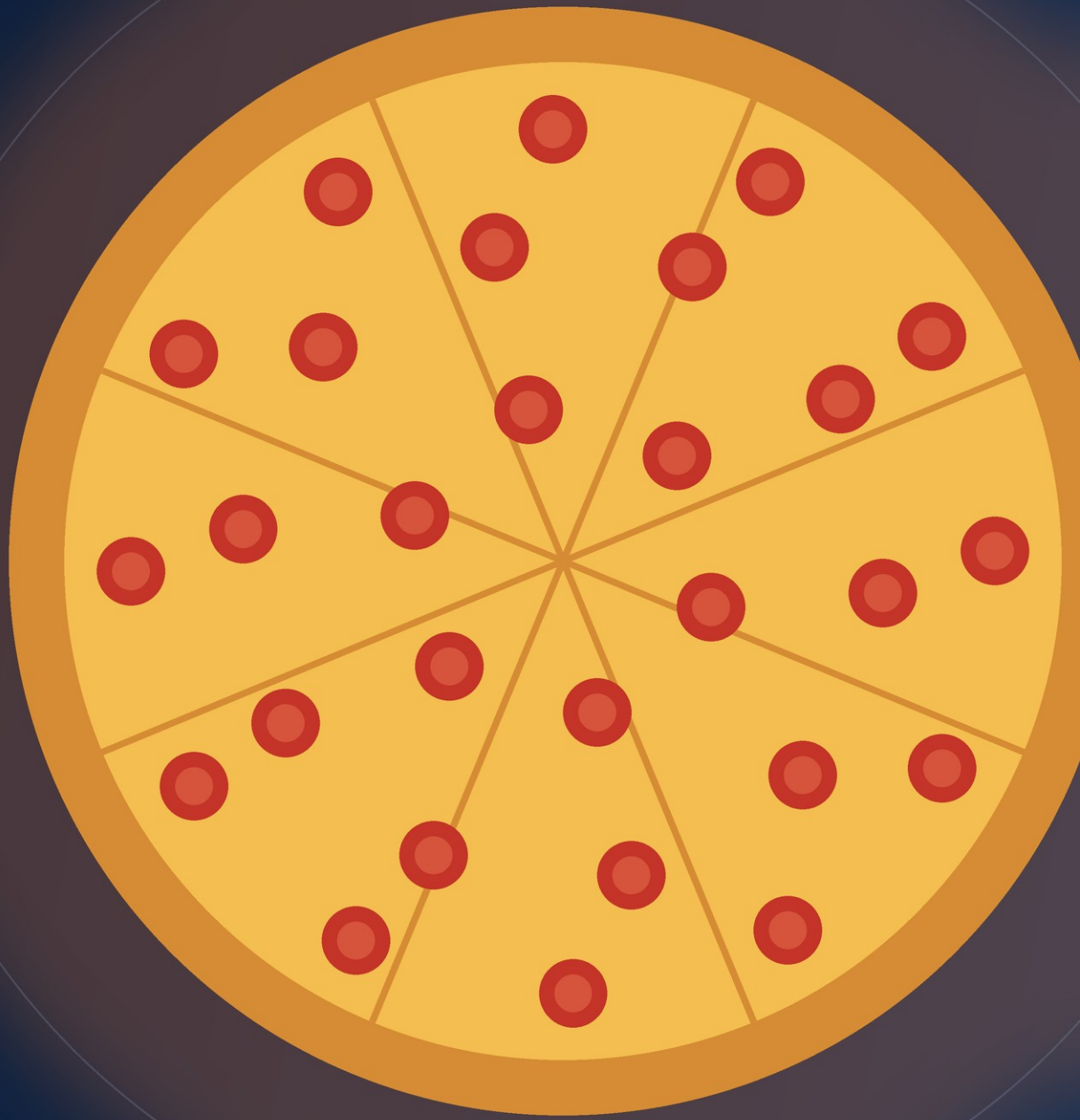
Pizza Sales Analysis

A MySQL project: revenue, orders and menu performance

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Tables: orders, order_details, pizzas, pizza_types

Concepts: JOINs, GROUP BY, subqueries, window functions



Q1. Calculate the total revenue generated from pizza sales

SQL File 1 | Limit to 1000 rows

```sql
-- calculated total revenue generated from pizza sales
SELECT
    ROUND(SUM(order_details.quantity * pizzas.price), 2) AS total_sales
FROM
    order_details
        JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	total_sales
▸	817860.05

Q2. Identify the highest-priced pizza

SQL File 2 | Limit to 1000 rows

```sql
-- identify the highest-priced pizza
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	name	price
▸	The Greek Pizza	35.95

Q3. Identify the most common pizza size ordered

SQL File 3 | Limit to 1000 rows

```sql
-- identify the most common pizza size ordered
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
        JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	size	order_count
▸	L	18526
	M	15385
	S	14137
	XL	544
	XXL	28

Q4. List the top 5 most ordered pizza types with quantities

SQL File 4 | Limit to 1000 rows

```sql
-- list the top 5 most ordered pizza types along with their quantities
SELECT pizza_types.name,
       SUM(order_details.quantity) AS quantity
FROM pizza_types
JOIN pizzas
    ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details
    ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	name	quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

Q5. Find the total quantity of each pizza category ordered

SQL File 5 | Limit to 1000 rows

```sql
-- find the total quantity of each pizza category ordered
SELECT pizza_types.category,
       SUM(order_details.quantity) AS quantity
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
JOIN order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	category	quantity
▸	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

Q6. Determine the distribution of orders by hour of the day

SQL File 6 | Limit to 1000 rows

```sql
-- determine the distribution of orders by hour of the day
SELECT
    HOUR(order_time) AS hour, COUNT(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	hour	order_count
▸	11	1231
	12	2520
	13	2455
	14	1472
	15	1468
	16	1920
	17	2336

	hour	order_count
▸	18	2399
	19	2009
	20	1642
	21	1198
	22	663
	23	28

Q7. Find the category-wise distribution of pizzas

SQL File 7 | Limit to 1000 rows

```sql
-- find the category-wise distribution of pizzas
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	category	count(name)
▸	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9

Q8. Calculate the average pizzas ordered per day

SQL File 8 | Limit to 1000 rows

```sql
-- group the orders by date and calculate the average number of pizzas ordered per day
SELECT
    ROUND(AVG(quantity), 0)
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	ROUND(AVG(quantity), 0)
▸	138

Q9. Find the top 3 pizza types by revenue

SQL File 9 | Limit to 1000 rows

```sql
-- find the top 3 pizza types based on revenue
SELECT pizza_types.name,
       SUM(order_details.quantity * pizzas.price) AS revenue
FROM pizza_types
JOIN pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
JOIN order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	name	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

Q10. Analyze the cumulative revenue generated over time

SQL File 10 | Limit to 1000 rows

```sql
-- calculate the cumulative revenue generated over time
SELECT order_date,
       SUM(revenue) OVER (ORDER BY order_date) AS cum_revenue
FROM
    (SELECT orders.order_date,
            SUM(order_details.quantity * pizzas.price) AS revenue
     FROM order_details JOIN pizzas
         ON order_details.pizza_id = pizzas.pizza_id
     JOIN orders
         ON orders.order_id = order_details.order_id
     GROUP BY orders.order_date) AS sales;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	order_date	cum_revenue
▶	2015-01-01	2713.8500000000004
	2015-01-02	5445.75
	2015-01-03	8108.15
	2015-01-04	9863.6
	2015-01-05	11929.55

Key Insights from the Analysis

1. Total revenue from pizza sales is 817,860.05, with about 138 pizzas ordered per day on average.

2. Large (L) is the most ordered size: 18,526 orders (38.1%). L, M and S together make up 98.8% of orders; XL (544) and XXL (28) are rare.

3. Top 5 pizzas by quantity are very close (2,371 to 2,453). The Classic Deluxe Pizza leads in quantity.

4. Revenue ranking differs from quantity: the Thai Chicken Pizza earns the most (43,434.25), followed by Barbecue Chicken (42,768) and California Chicken (41,409.5).

5. The Classic category sells the most pizzas (14,888, 30.0%) even though it has fewer pizza types (8) than Supreme and Veggie (9 each).

6. Orders peak at 12 PM to 1 PM (2,520 and 2,455) and 5 PM to 6 PM (2,336 and 2,399); these four hours cover about 45% of the orders by hour.

7. The highest-priced pizza is The Greek Pizza at 35.95; cumulative revenue grows steadily day by day.